Open-Source EDA as a Strategic Alternative for European Chip Design

Export-control risk at the design-tool layer

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What changed: access is no longer only a purchasing issue

For critical technologies, tool access, export controls and sanctions can change the supply chain.

Old engineering question		Question companies now need to ask
Can we buy the tool?	→	Are we allowed to use it for this customer and destination?
Can we tape out at this fab?	→	Does the full route add export-control exposure?
Is the dependency acceptable?	→	What would happened to the product if access changes?

For chip design, the risk can appear before fabrication, at the design-tool layer.

FDPR, simplified

A chip made outside the United States can still fall under US export rules.

FDPR = Foreign Direct Product Rules

Dependency trace from a controlled input to a foreign-made output and a restricted transaction.

1 controlled input		2 foreign-made output		3 restricted transaction
specified technology or software, or a plant or major component derived from it	AND	the chip is produced from that controlled input	AND	destination, listed party, end user or end use matches the rule

Conceptual model

If all three conditions line up, the chip may need a licence or shipment may be blocked.

Why EDA matters

An EDA tool can be part of the controlled input in step 1. The customer, destination and use case decide whether step 3 is triggered.

Important limit: using a US EDA tool is not enough by itself. FDPR is a conditional rule, not an automatic label.

Why engineers should care

Export control can interrupt a product even when the design is technically complete.

$95M

BIS administrative penalty against Cadence in 2025

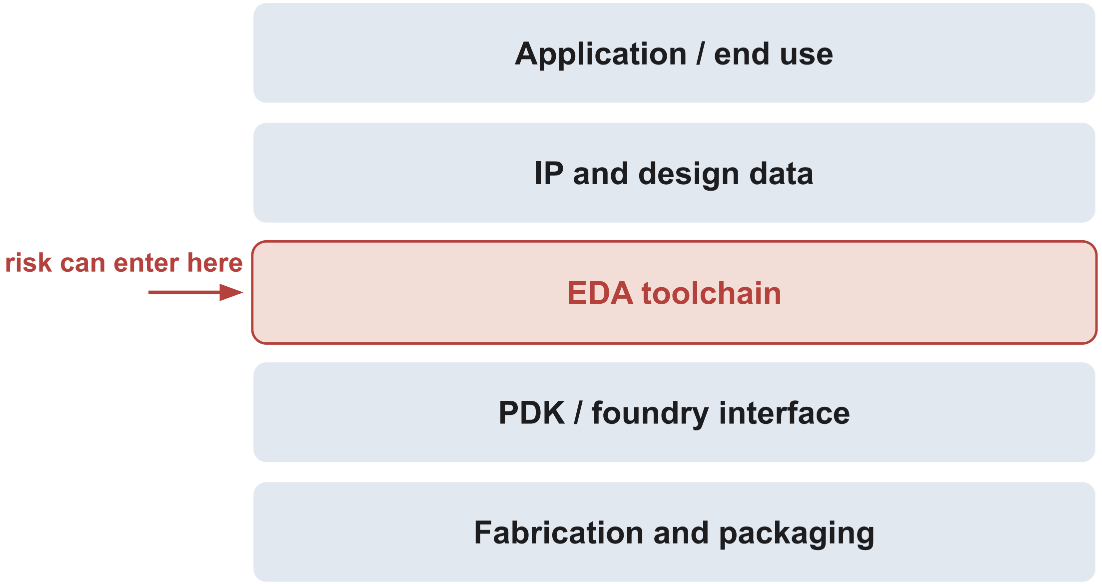
The case concerned unauthorized exports of EDA hardware, software and design technology. It was not an FDPR ruling.

1	**Tool or licence access changes**	The team may be unable to rerun implementation or signoff.
2	**Shipment or customer is blocked**	Milestones and contracts move immediately.
3	**Redesign and requalification**	ASIC recovery can take months or years.

EDA is already inside export-control enforcement. For engineers, the consequence is loss of continuity.

The EDA toolchain is part of the dependency graph

Before fabrication, a chip is simulated, verified, synthesized, placed, routed, and signed off. Using those tools can create additional legal dependencies to foreign export-control law.



Open-source EDA can change one node in this graph. It does not remove the others.

"Published" Software Exemptions

Software that qualifies as "published" or "publicly available" operates under regulatory exemptions:

EAR (15 C.F.R. § 734.7): Under the US Export Administration Regulations, software or technology that has been made publicly available without restrictions on its further dissemination is considered "published" and is generally not subject to the EAR.

Exceptions: *Certain categories remain subject to the EAR, including published encryption software classified under ECCN 5D002 (§ 734.7(b)) and 3D printing files for firearms (§ 734.7(c)).*

EU Dual-Use Regulation (Regulation (EU) 2021/821, Annex I – General Software Note): The EU Dual-Use Control List does not control software that is "generally available to the public" or "in the public domain".

Exceptions: *The General Software Note applies only to Categories 1–9 and does not apply to Category 5, Part 2 ("Information Security").*

One link removed

If the EDA tool is published and outside the applicable controls, that tool does not add export-control exposure to the chip.

Published open-source EDA → **IC design** → **Fabrication in another country** → **Product, customer and end use**

The design-tool link is reduced because the published software is outside the relevant control scope.

Remaining dependencies still need analysis

- PDK and standard cells
- Third-party IP
- Cloud and compute services
- Foundry equipment and process technology
- Destination, end user and end use

Where this can work first

Start where open flows are technically credible and the validation problem is bounded.

Mature-node digital blocks

180 nm, 130 nm and 110 nm flows are a realistic entry point because designs are smaller, rules are more stable and commercial fallback is available.

Industrial evidence

Examples already exist: a reported ChipFlow 130 nm BCD test chip and a DECTRIS 110 nm serial-link block.

Support infrastructure

Simulation, CI, waveform tools, cocotb and mixed-signal support can be adopted before replacing signoff.

Hard part: PDK translation

The bottleneck is often not RTL. It is LEF, Liberty, layers, DRC assumptions, macros, corners, parasitics and signoff correlation.

Best first target: bounded digital blocks where failures are visible and recovery is cheap.

A adoption path for companies

The objective is not ideological purity. It is to reduce dependence on a single toolchain and keep a workable alternative available.

1 → 2 → 3 → 4 → 5

1. **shadow flow**
   Run open tools beside the commercial flow
2. **non-critical blocks**
   Use for test chips, wrappers, control logic
3. **reproducible evidence**
   Track QoR, failures, runtime, constraints
4. **approved use cases**
   Define where the toolchain is acceptable
5. **compliance record**
   Document publication status and dependencies

What the community can do now

Treat open EDA as continuity infrastructure, not only as a cheaper tool.

For maintainers and researchers

Publish complete flows	Include scripts, constraints and tool versions.
Track dependencies	Record PDK, IP, cloud and external services.
Report failures openly	Benchmarks need negative results and reproducibility.

For companies and funders

Fund maintenance	Validation and support matter more than one-off prototypes.
Define approved use cases	Start with bounded industrial workloads.
Connect engineering to policy	Make risk reduction visible in procurement and strategy.

Open EDA cannot remove every export-control risk. It can remove one dependency we can control.

Thank you

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